

Gold-Free Faithfulness Metrics for NL-to-FOL Translation

Framing

Translating natural language into first-order logic (NL-FOL) is a prerequisite for neurosymbolic reasoning pipelines [10], yet evaluating whether a candidate formula faithfully captures a sentence's meaning remains an open problem. Gold FOL annotations are expensive, non-unique, and unreliable: Brunello et al. [3] found that approximately 42% of entries in both FOLIO [1] and MALLS [2] contain incorrect formalisations. Existing metrics either require a gold formula (exact match, prover-checked equivalence) or only verify that the output parses [5, 6]. The user's own pilot study used structural consistency checks (arity agreement, predicate-set overlap across reruns, joint satisfiability) that measure stability but not whether a formula says what its sentence says.

This run investigates gold-free metrics that predict faithfulness to the text given only a sentence and a candidate formula, with no reference formula, no ontology, and no domain knowledge. The evaluation is a meta-evaluation: we build a labelled set of (sentence, formula) pairs from public benchmarks whose gold has been audited, and measure each metric's ability to discriminate correct translations from errors.

Iteration 1

1.1 Strategy

The iteration screens four candidate metric families head-to-head on a frozen evaluation set, with a disguised cheap LLM judge as the bar to beat.

Candidates. (A) FOL-Triage, a three-layer cascade: L1 (formula lint: deterministic formula smells), L2 (text accounting: bag-of-words and role-aware coverage), and L3 (role questionnaire: a small LLM reads the text and z3 reads the formula; their structured answers are compared). (B) Round-trip verification: verbalise the formula back to natural language and compare to the original via NLI or embedding similarity [7]. (C) Cross-system consensus: collect translations of the same sentence from multiple independent systems, measure z3-equivalence between the candidate and the peer pool, and score disagreement [8]. (D) Disguised cheap judge and baselines: an LLM judge, structural pilot metrics, and round-trip features evaluated on nonce-disguised sentences to control for memorisation.

Evaluation set. The frozen screen uses Logic-LM [10] released outputs on FOLIO validation: three LLM systems (GPT-3.5-turbo, GPT-4, text-davinci-003) producing 796 items. Track L contains LLM outputs labelled by z3 equivalence to corrected gold [4]; track H contains the 294 original human gold formulas, of which 57 are errors according to the corrected labels. The parallel run (run_qY2a2IS-WLIs) measured 43.9% of solver-non-equivalent candidates as faithful by a three-family panel; iteration 2 will re-score with those adjudicated labels.

Decision rule. A candidate must achieve $\text{AUROC} \geq 0.65$ on track L (ERROR vs CORRECT), with coverage $\geq 90\%$, rewrite false-alarm rate $\leq 10\%$ per rewrite family, and cost $\leq \$0.002$ per item. It advances only if its AUROC exceeds the disguised cheap judge's, with a paired bootstrap CI excluding zero.

Held-out dataset (Artifact E). A separate 700-sentence, 8,507-row held-out set was constructed from FOLIO training premises, MALLS training long/conditional/exception strata, and FOLIO curated conclusions, each translated by 9 LLM families plus ccg2lambda plus GPT-4 gold. Labels were assigned by z3 equivalence to audited gold, with contested cases adjudicated by a three-member panel (Claude Haiku-4.5, GLM-4.6, Kimi-K2). This set is reserved for iteration 2 confirmation.

The dataset contains 1,750 CORRECT, 5,005 ERROR, 1,086 UNPARSEABLE, 276 UNRESOLVED, and 390 CONTESTED items. The panel calibration subset (173 items) shows that the panel agrees with expert-corrected labels on both faithful and unfaithful items. A screen audit of 1,173 items from the Logic-LM screen confirms the label assignment. The dataset cost \$9.83 in API calls.

1.2 Experiment A: FOL-Triage

[ARTIFACT:art_d0njuqy2Csj-]

FOL-Triage was evaluated on the frozen screen with Gemini 2.5 Flash Lite as the primary LLM for L3. Total API spend was \$0.29.

Headline AUROCs (track L, ERROR vs CORRECT, n = 477, 199 errors)

Metric	AUROC	95% CI	AUPRC
L1 (any smell)	0.508	[0.499, 0.519]	0.425
L2 bag-of-words	0.737	[0.677, 0.801]	0.690
L2 role-aware	0.557	[0.527, 0.594]	0.476
L3 score	0.756	[0.701, 0.809]	0.657
Fused (fitted on H, no leak)	0.759	[0.695, 0.825]	0.740
Fused (all H)	0.827	[0.778, 0.874]	0.771
Fused (cross-fit L)	0.864	[0.824, 0.902]	0.789
Decomposed judge (LLM reads formula)	0.716	[0.645, 0.789]	0.717
L3 with local Qwen2.5-1.5B	0.708	[0.637, 0.774]	0.603

The primary no-leak fused score is 0.759. The fused-all-H and cross-fit-L numbers (0.827, 0.864) are shown for completeness but are not valid held-out estimates: the former uses the full track H for fitting, and the latter is a within-L cross-fit that overstates generalisation.

Track H (human gold errors, n = 258, 57 errors). Fused out-of-fold AUROC is 0.777 [0.698, 0.847]. L1 alone reaches 0.655, compared to 0.508 on track L. The lint rules fire on human annotation errors but not on LLM outputs, because the two populations produce different formula smells.

L1 lint: a negative result on LLM outputs

L1 flags formulas with deterministic smells (existential implication, free variables, arity clashes, vacuity, trivial formulas, dangling predicates). On track L, the true positive rate is 2.0% at a false positive rate of 0.4%, giving AUROC 0.508. This is a negative result: L1 lint does not transfer from human annotation errors (where it is effective, AUROC 0.655) to LLM outputs. The reason is that LLM-generated formulas rarely contain the syntactic pathologies L1 detects; their errors are semantic (wrong predicates, missing conditions, scope mistakes).

All nine individual smell rules have false-alarm rates below 3% on every system:

Smell	gpt-3.5 FA	gpt-4 FA	davinci-003 FA
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EX_IMP	0.0	0.0	0.0
ALL_AND	0.0	0.0	0.0
IFF_RESTR	0.0	0.009	0.0
GLUE	0.0	0.0	0.0
FREE	0.0	0.0	0.0
VACUOUS	0.0	0.0	0.0
TRIVIAL	0.0	0.0	0.0
ARITY	0.0	0.0	0.0
DANGLING	0.0	0.0	0.0

Only the FREE smell fires on any track-L errors (4 items).

L2 role-aware check: below bag-of-words

The role-aware UD-based check (L2-role) achieves AUROC 0.557, which is 0.18 below the simpler bag-of-words check (L2-bow, 0.737). The role-aware layer adds no discriminative power over word overlap.

Decomposed judge ablation

Replacing z3 with an LLM that reads the formula directly (the decomposed judge) drops L3 AUROC from 0.756 to 0.697 (Δ AUROC = -0.059). The LLM gets only 57.6% of binary argument slots correct on track L and 36.4% on track H. The z3 answerer is significantly more accurate for structured formula reading.

Per-type sensitivity (track L, at binary threshold)

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	5	0.0	0.2	1.0	0.2	0.4	1.0
QUANT	2	0.0	0.0	0.5	0.0	0.5	0.5
CONN	4	0.0	0.0	0.0	0.0	0.0	0.0
DROP	142	0.028	0.479	0.127	0.197	0.599	0.606
ADD	150	0.020	0.473	0.107	0.173	0.653	0.573
SWAP	12	0.0	0.0	0.083	0.25	0.083	0.333
BIND	8	0.125	0.0	0.125	0.0	0.375	0.25
COMPOUND	187	0.118	0.428	0.166	0.160	0.599	0.578

FOL-Triage is most sensitive to ADD and DROP errors (the largest categories), with the fused score catching about 60%. It misses CONN errors entirely (0/4) and detects SWAP poorly (1/12).

Negation errors are fully caught by the cascade (which fires L2-role, where monotonicity shifts are visible) but L1 and L2-bow miss them.

On track H (57 errors), the per-type pattern differs. UNGLUE errors (n = 5) are caught at 100% by L1 and the fused score, because they produce detectable formula smells.

Per-type sensitivity (track H)

Error type	n	L1	L2-bow	L2-role	L3	Fused	Cascade
NEG	4	0.0	0.0	0.25	0.0	0.25	0.25
REV	1	0.0	0.0	0.0	0.0	0.0	0.0
QUANT	1	0.0	0.0	0.0	0.0	0.0	0.0

RESTR	1	1.0	1.0	0.0	0.0	1.0	1.0
CONN	8	0.25	0.125	0.0	0.375	0.375	0.375
MOVE	2	0.5	0.0	0.5	0.0	0.5	1.0
DROP	14	0.143	0.357	0.0	0.143	0.357	0.5
ADD	19	0.158	0.368	0.053	0.211	0.526	0.579
SWAP	2	0.0	0.5	0.0	0.5	0.5	0.5
BIND	1	1.0	1.0	0.0	0.0	1.0	1.0
UNGLUE	5	1.0	0.4	0.0	0.2	1.0	1.0
COMPOUND	33	0.303	0.182	0.242	0.182	0.545	0.636

Invariance under meaning-preserving rewrites

Five rewrite families were tested on 150 track-L CORRECT items. The table shows the false-alarm rate on the rewritten formula and the flip rate (fraction of items whose binary verdict changes):

Rewrite	n	L1 applies	L1 FA	L1 flip	bow FA	bow flip	role FA	role flip	L3 FA	L3 flip	Fused FA	Fused flip
RENAME	150		0.0	0.0	0.18	0.093	0.013	0.0	0.033	0.007	0.22	0.013
REORDER	34		0.0	0.0	0.029	0.0	0.059	0.0	0.059	0.0	0.235	0.0
DEMORGAN	33		0.0	0.0	0.030	0.0	0.061	0.0	0.030	0.030	0.182	0.061
CONTRAPOSITIVE	40		0.025	0.025	0.10	0.0	0.025	0.025	0.025	0.0	0.275	0.0
PRENEX	1		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

The rewrite false-alarm gate ($\leq 10\%$ per family) fails. The worst-family fused FA is 0.275 (CONTRAPOSITIVE), but this equals the FA on the original formulas; the flip rate is at most 0.061. The high FA is not caused by the rewrites but by the fused score's calibration: it was fitted on track H where the CORRECT base rate is higher, producing a systematic over-flagging on track L (FA on track-L CORRECT items is 0.194 versus the 0.10 calibration target).

L1, L2-role, and L3 individually pass the 10% gate. L2-bow fails only under RENAME (FA = 0.18), because renaming predicates breaks the bag-of-words match between text and formula.

[Figure fig_rewrite_fa: generated in the report phase, not in this draft]

Coverage

Track	n	Parseable	Full coverage rate
L	753 693		0.920
H	302 279		0.924

Parse rates by system: gpt-3.5-turbo 0.889, gpt-4 0.946, text-davinci-003 0.921. Unparseable outputs are scored as $p = 1$ (maximum error probability) by the fused metric, so all items receive a score. The coverage gate ($\geq 90\%$) passes.

Cost

L1 and L2 are free (CPU-only). L3 costs \$0.175 per 1,000 calls with Gemini 2.5 Flash Lite. The total per-item cost is \$0.0002, passing the \$0.002 gate. CPU time (L1 + L2 + z3): median 35 ms, P95 90 ms. L3 latency: median 630 ms, P95 1.68 s.

Contamination check

L3 was run on nonce-disguised sentences (content words replaced by nonsense words) on the full

track L. AUROC with exact text-formula alignment: 0.765; with disguised text: 0.770. Paired difference: -0.005 [$-0.063, 0.049$]. There is no evidence that the LLM benefits from having seen the public benchmark sentences.

Complexity strata (track L, fused score)

Stratum	n (error/correct)	Testable	Fused AUROC	L2-bow AUROC	L3 AUROC
words < 12	443 (179/264)	Yes	0.781	0.742	0.772
words 12-19	31 (18/13)	No	0.521	0.596	0.541
words $\geq$ 20	3 (2/1)	No	0.500	1.000	0.000
0 quantifiers	302 (130/172)	Yes	0.743	0.728	0.771
1 quantifier	164 (65/99)	Yes	0.794	0.748	0.721
$\geq$ 2 quantifiers	11 (4/7)	No	0.750	0.768	0.607
depth $\leq$ 1	296 (124/172)	Yes	0.762	0.745	0.779
depth 2-3	173 (73/100)	Yes	0.756	0.717	0.710
depth $\geq$ 4	8 (2/6)	No	0.667	0.750	0.500
0 conditions	344 (145/199)	Yes	0.775	0.759	0.791
1-2 conditions	131 (54/77)	Yes	0.716	0.677	0.628

The frozen screen is dominated by short, simple sentences (93% under 12 words). The long and complex strata (12+ words, 2+ quantifiers, 3+ conditions) have too few items to test. This limitation motivates the held-out dataset (Artifact E), which oversamples these strata.

Per-system AUROC (descriptive, n = 3 systems)

System	n	Error rate	Fused AUROC	L1 AUROC	L2-bow AUROC	L3 AUROC
gpt-3.5-turbo	143	0.378	0.689	0.500	0.650	0.722
gpt-4	177	0.367	0.700	0.496	0.716	0.720
text-davinci-003	157	0.510	0.857	0.525	0.811	0.817

FOL-Triage is most discriminative on text-davinci-003, the weakest system. The Kendall τ between system error rates and mean fused scores is 0.333 (3 systems, descriptive only).

Label quality observations

The track-L label distribution reveals a systematic issue: 58.8% of typed ERROR items have only ADD + DROP repair operators. On inspection, most of these are vocabulary or arity choices (e.g. Lunch(james) vs HasLunch(james, company)) that the vocabulary aligner cannot bridge. These are likely CORRECT under a looser equivalence definition. The track-H original gold has 27.1% error rate (errors, uncertain, and reading-choice items) and 5.8% unparseable rate. Among non-equivalent parseable items, 21.9% are correct-but-not-equivalent (VOCAB or GRAN class).

1.3 Experiment C: Cross-System Consensus

[ARTIFACT:art_i3cVDxBp-USk]

The consensus metric collects translations of the same sentence from multiple independent LLM families, checks pairwise z3 equivalence, and scores each candidate by its disagreement with the pool. Total API spend was \$1.52.

Headline AUROCs (track L, ERROR vs CORRECT, n = 546, 249 errors)

Metric	AUROC	95% CI	AUPRC
c_score (1 – eq_frac, all peers)	0.866	[0.821, 0.906]	0.827
c_score (6 fresh peers only)	0.872	[0.830, 0.911]	0.825
c_score (non-OpenAI peers)	0.863	[0.820, 0.903]	0.808
c_score (2 Logic-LM peers only)	0.753	[0.701, 0.806]	0.659
medoid depth	0.722	[0.673, 0.774]	0.622
cluster entropy	0.773	[0.724, 0.820]	0.691
SC-5 cheap (gpt-4.1-nano)	0.725	[0.669, 0.781]	0.627
SC-5 same model	0.716	[0.632, 0.796]	0.628
SC-5 entropy	0.631	[0.568, 0.693]	0.539
Prediction set instability (SC)	0.688	[0.626, 0.750]	0.602
Prediction set instability (pool)	0.770	[0.717, 0.821]	0.721
FOL length	0.527	[0.470, 0.586]	0.485
Misalignment (vocab)	0.835	[0.794, 0.876]	0.821

[Figure fig_consensus_auroc: generated in the report phase, not in this draft]

The c_score achieves the highest AUROC in the iteration at 0.866. It beats SC-5 (the self-consistency baseline) by $\Delta\text{AUROC} = +0.140$ [0.085, 0.197]. Cross-family consensus also exceeds sampling self-consistency in the related SAC3 framework [17], which uses semantic-aware cross-check consistency for hallucination detection in black-box LLMs. Using only the six fresh peer translations (not the three Logic-LM systems) gives 0.872, slightly higher than using all peers. Restricting to the two other Logic-LM systems drops the AUROC to 0.753, showing that diversity across model families is essential.

The misalignment score (0.835) is high because CORRECT items are by construction alignable to the gold in the labeller, so misalignment alone is a strong proxy for error. This does not generalise to settings without a shared vocabulary.

RENAME rewrite invariance: failure

Under predicate renaming (replacing predicate names with WordNet synonyms or nonce names), the c_score false-alarm rate is 96% (100% for constant-only renaming). This is expected and fundamental: two formulas with different predicate names are not z3-equivalent, so any rename breaks the equivalence check. The consensus metric is invariant to rewrites that preserve z3 equivalence (reordering, De Morgan, contrapositive) but not to vocabulary changes.

Blind-spot analysis

38% of ERROR items are z3-equivalent to the medoid (the most popular peer formula). This places a ceiling on recall of approximately 62%. The blind-spot rate is 5% for polarity errors and 57% for structural errors.

Cost

The peer pool costs \$0.0015 per sentence (6 peers via API). Total experiment spend: \$1.52. z3 pairwise equivalence checking is free.

1.4 Experiment D: Disguised Judge and Baselines

This experiment establishes the bar: every baseline metric and every judge variant on the same shared screen, scored with nonce-disguised sentences to control for memorisation. Total API spend was \$2.02 (after the OpenRouter key was restored at 15:11 UTC; local judges ran during the outage and are kept as secondary rows).

The bar: disguised cheap judge

The primary cheap judge (Gemini 2.5 Flash Lite, JSON 0-100 scoring, nonce-disguised sentences) achieves AUROC 0.777 [0.720, 0.827] on track L (n = 524, 230 errors, 294 correct).

Full baseline table (track L, ERROR vs CORRECT, n = 524)

Metric	AUROC	95% CI
parse_fail	0.500	[0.500, 0.500]
pilot_joint_conflict	0.513	[0.488, 0.539]
pilot_arity_incons	0.500	[0.500, 0.500]
pilot_shape_incons	0.524	[0.470, 0.578]
pilot_dangling	0.533	[0.474, 0.592]
pilot_undeclared	0.525	[0.510, 0.542]
pilot_rerun_jacc	0.720	[0.659, 0.780]
rt_nli_min	0.710	[0.648, 0.769]
rt_nli_fwd	0.652	[0.588, 0.713]
rt_nli_bwd	0.681	[0.621, 0.742]
rt_nli_contra	0.628	[0.566, 0.686]
rt_nli_min_alt	0.689	[0.627, 0.749]
rt_embed_cos	0.633	[0.574, 0.696]
rt_reformalise_eq	0.513	[0.479, 0.547]
judge_cheap_orig	0.749	[0.695, 0.801]
judge_cheap_disg	0.777	[0.720, 0.827]
judge_cheap2_orig (gpt-4.1-nano)	0.750	[0.691, 0.810]
judge_cheap2_disg	0.714	[0.648, 0.777]
judge_strong_orig (gemini-3.1-pro)	0.802	[0.715, 0.877]
judge_strong_disg	0.730	[0.655, 0.803]
judge_local_qwen8b_disg	0.757	[0.702, 0.811]
judge_local_llama8b_disg	0.671	[0.604, 0.737]
judge_local_qwen14b_disg	0.760	[0.681, 0.834]
rt_nli_min_localverb	0.740	[0.680, 0.796]

[Figure fig baseline_auroc: generated in the report phase, not in this draft]

Pilot structural metrics: negative result

The user's pilot structural metrics are null on this screen. Joint conflict achieves AUROC 0.513, arity inconsistency 0.500, shape inconsistency 0.524, dangling predicates 0.533, and undeclared predicates 0.525. The cross-system predicate-set Jaccard (pilot_rerun_jacc) reaches 0.720, the only pilot metric with signal, but this is a proxy for the consensus approach tested in Experiment C.

Round-trip verification

Round-trip NLI (verbalise the formula, compare to the original sentence via bidirectional NLI entailment) achieves AUROC 0.710 [0.648, 0.769]. This is 0.067 below the cheap judge ($p = 0.008$ by DeLong test). The round-trip reformalisation check (verbalise, re-translate to FOL, check z3 equivalence with the original) achieves only 0.513, because 76% of items tie at the same score.

Round-trip NLI using local verbalisation (the LLM verbalises on the local GPU instead of the API) reaches 0.740 [0.680, 0.796], slightly higher than the API version.

Frontier judge

The strong judge (Gemini 3.1 Pro Preview) achieves AUROC 0.802 [0.715, 0.877] on the strong subsample ($n = 291$). Under disguise it drops to 0.730 [0.655, 0.803]. The Δ AUROC over the cheap judge is $+0.024$ [$-0.075, 0.111$], not significant.

Cross-fitted baseline combinations (track L)

Combination	AUROC	95% CI
S1 (structural)	0.698	[0.634, 0.761]
S2 (round-trip)	0.687	[0.625, 0.749]
S3 (judges disguised)	0.771	[0.705, 0.831]
S4 (all cheap features)	0.817	[0.762, 0.867]
S6 (S4 + local judges)	0.810	[0.753, 0.861]

S4 (all cheap features combined via cross-fitted logistic regression) achieves 0.817. This is $+0.039$ [$-0.010, 0.087$] over the cheap judge alone, not significant, but it sets the empirical ceiling for cheap baselines.

Contamination analysis

No evidence of contamination was found for any judge. The difference-in-differences (DiD) test compares the disguise effect on track H (where the LLM could have memorised public gold) to track L (where it evaluates its own outputs, not memorisable). All DiD CIs include zero:

Judge	DiD (H – L)	95% CI
flash-lite	-0.042	[$-0.170, 0.094$]
gpt-4.1-nano	-0.077	[$-0.188, 0.023$]
gemini-3.1-pro	$+0.002$	[$-0.115, 0.104$]
Qwen3-8B	-0.045	[$-0.161, 0.079$]
Llama-3.1-8B	$+0.023$	[$-0.112, 0.157$]
Qwen3-14B	-0.073	[$-0.195, 0.056$]

The gold-recall probe confirms this: when the primary model translates track-H sentences from scratch, it matches the corrected gold 15.9% of the time but the original (public) gold only 4.4%. Predicate-name Jaccard with the original gold minus Jaccard with the Logic-LM GPT-4 output is -0.068 [$-0.171, 0.034$]; the model is not reproducing the public gold.

Invariance of judges under meaning-preserving rewrites

Judges are highly sensitive to meaning-preserving rewrites. The FA on CONTRAPOSITIVE rewrites is 0.67 for the cheap judge (original), 0.51 for the cheap judge (disguised), and 0.95 for the cheap2 judge. This compares poorly to FOL-Triage's fused flip rate of 0.0 on CONTRAPOSITIVE.

The full invariance table (track L CORRECT items, $n = 270$):

Metric	RENAME flip	REORDER flip	DEMORGAN flip	CONTRA flip	ALL flip
judge_cheap_orig	0.27	0.12	0.23	0.64	0.29
judge_cheap_disg	0.44	0.12	0.46	0.41	0.39
judge_cheap2_orig	0.49	0.12	0.46	0.95	0.49
rt_nli_min	0.37	0.10	0.31	0.33	0.31
rt_embed_cos	0.57	0.07	0.44	0.38	0.44
FOL-Triage fused	0.013	0.0	0.061	0.0	0.013

FOL-Triage has the lowest flip rate of any metric tested. Judges and round-trip methods change their verdict on 29-49% of meaning-preserving rewrites.

Per-error-type sensitivity of judges (track H, curator view, $n = 46$ errors)

Group	n	judge_cheap_orig	judge_cheap_disg	judge_cheap2_orig	rt_nli_min
1-op errors	13	0.308	0.538	0.462	0.231
2-op errors	11	0.545	0.818	0.727	0.273
ADD	12	0.500	0.750	0.667	0.250
COMPOUND	22	0.545	0.864	0.773	0.273
DROP	11	0.455	0.636	0.455	0.182
CONN	3	0.667	0.667	1.000	0.333
NEG	2	0.500	0.500	1.000	0.500
SWAP	2	0.000	1.000	0.500	0.000

The disguised judge is more sensitive than the original judge for most error types, suggesting that nonce disguise forces the model to read more carefully.

Judge error-type classification accuracy

The judge's ability to name the correct error type is poor. On track-H 1-op errors, it identifies the correct repair operator 0% of the time (0/12). On 2-op errors, it is correct 45% (5/11). On track L, accuracy is 20% for 1-op and 27% for 2-op errors.

Complexity analysis (track L)

A logistic regression of judge correctness on standardised word count yields a coefficient of -0.48 (SE 0.11, $p < 0.001$): the judge is significantly less reliable on longer sentences. On track H (where longer sentences are available), the coefficient is -0.14 (SE 0.17, $p = 0.41$), consistent in direction but underpowered.

Stratum	n (err/cor)	judge_cheap_disg AUROC	rt_nli_min AUROC
words < 12	485 (210/275)	0.784	0.713
words 12-19	36 (18/18)	0.667	0.670
quantifiers 0-1	512 (226/286)	0.784	0.712
depth ≤ 3	466 (199/267)	0.781	0.702
depth 4-5	54 (30/24)	0.762	0.828
0 conditions	394 (182/212)	0.783	0.745
1-2 conditions	128 (48/80)	0.765	0.653

1. No contamination was detected for any LLM-based metric.

What remains for iteration 2: (a) Re-evaluate all candidates under adjudicated labels from Artifact E. (b) Compute the paired Δ AUROC between FOL-Triage and the disguised judge on shared items. (c) Test whether combining the consensus metric with FOL-Triage or the judge produces a significant improvement. (d) Confirm on the held-out set.

Iteration 2

2.1 Strategy

Iteration 2 has three goals: (i) confirm iteration 1's ordering on the held-out dataset E under both the solver labels (R_AB, R_A) and panel-adjudicated labels (R_ADJ); (ii) test whether fusing the consensus signal (PEER) with text-based checks (TEXT = L2-bow + L3) improves over each alone; and (iii) construct a typed perturbation suite (PERTURB) for controlled sensitivity testing.

The pre-registered bar for this iteration is the flash-lite judge on disguised sentences, with the local Qwen3-8B judge as fallback if the API is unavailable. The shared OpenRouter key hit its daily limit at 18:00 UTC on 2026-09-23, before the held-out experiment completed, and did not recover before the run deadline. The flash-lite bar is therefore UNTESTABLE on E; the fallback local judge (Qwen3-8B, rubric B, greedy; screen AUROC 0.757 vs 0.777 for flash-lite) is the labelled secondary bar for all rows reported below.

2.2 Experiment 5: PEER + TEXT on held-out dataset E

PEER + TEXT is a logistic fusion of the consensus signal (c_score with shared-vocabulary alignment, denoted c_score_align) and the text-based FOL-Triage layers (L2-bow, L3 questionnaire). The fusion weights were fitted on the iteration-1 screen and frozen before any E score was computed (prereq sha256 b9f28b1bf6bf). API spend was \$0 (all components ran on CPU and the local GPU; no OpenRouter calls completed).

Name-free alignment: failed the pre-registered screen gate

A name-free variant of the consensus metric (graded clause consensus, g_score) was pre-registered as the primary PEER signal. It matches formula clauses by structural role without relying on shared predicate names, using anchored-cost assignment and Murty k-best matching. Under the pre-registered selection rule, both name-free variants (NF-pure and NF-anchored) failed the screen gate: for both, more than 10% of AGREE track-L CORRECT items tie at $g = 1$, making the deterministic false-alarm threshold degenerate (ROLE_PERMUTE recall = 0).

A post-hoc analysis with fractional tie-breaking showed this failure was a threshold-tie artefact. NF-anchored passes both gates under fractional scoring (RENAME FA 0.074, ROLE_PERMUTE recall 0.771); NF-pure still fails ROLE_PERMUTE (recall 0.186). The NF-anchored fusion was fitted post-hoc and applied to E as a labelled sensitivity analysis. On E (R_AB pooled), the NF-anchored fusion achieves AUROC 0.759 [0.72, 0.79], which is 0.031 below the pre-registered PEER + TEXT fusion (0.790), with the CI excluding zero [−0.042, −0.020]. The shared-aligner confound with the solver labeller remains open.

Headline AUROCs on E (R_AB, ERROR vs CORRECT)

Subset	n (err/cor)	PEER + TEXT	c_score_align	NF c_score	L2-bow	L3	Local judge
R_AB pooled	2686 (1822/864)	0.790 [0.75, 0.82]	0.782 [0.74, 0.82]	0.743 [0.70, 0.78]	0.734 [0.71, 0.76]	0.579 [0.53, 0.62]	0.712
R_AB L25	873 (697/176)	0.755 [0.70, 0.80]	0.712 [0.65, 0.77]	0.651 [0.59, 0.71]	0.722 [0.67, 0.77]	0.560 [0.49, 0.63]	0.664
R_AB L20	821 (592/229)	0.747 [0.69, 0.80]	0.722 [0.66, 0.78]	0.643 [0.58, 0.70]	0.715 [0.66, 0.76]	0.547 [0.48, 0.62]	0.683
R_AB EXC	606 (384/222)	0.777 [0.72, 0.83]	0.814 [0.76, 0.86]	0.782 [0.72, 0.84]	0.669 [0.61, 0.72]	0.620 [0.55, 0.69]	0.663
R_AB CTRL	386 (149/237)	0.708 [0.57, 0.84]	0.742 [0.62, 0.86]	0.741 [0.61, 0.86]	0.613 [0.54, 0.72]	0.490 [0.37, 0.62]	0.739
R_AB long pool	2300 (1673/627)	0.768 [0.74, 0.80]	0.759 [0.72, 0.79]	0.704 [0.67, 0.74]	0.717 [0.69, 0.75]	0.580 [0.54, 0.62]	0.653
R_A pooled L20 + EXC	449 (261/188)	0.818 [0.73, 0.88]	0.860 [0.79, 0.92]	0.802 [0.72, 0.87]	0.678 [0.61, 0.74]	0.587 [0.51, 0.66]	0.554

[Figure fig_held_out_auroc: generated in the report phase, not in this draft]

PEER + TEXT vs the local judge

The paired Δ AUROC of PEER + TEXT minus the local judge is +0.078 [0.043, 0.112] on R_AB pooled (DeLong $p < 0.0001$). On the long pool (L25 + L20 + EXC), the gap widens to +0.115 [0.075, 0.157]. On R_A (tier-A adjudicated labels, L20 + EXC), the gap is +0.264 [0.157, 0.359].

The flash-lite judge comparison is untestable: only 20 items received flash-lite scores before the key was exhausted, yielding a Δ AUROC of +0.057 [−0.200, 0.360] ($n = 20$, uninformative).

Does fusion beat c_score_align alone?

The stratified AUROC (pairing only within complexity strata, removing the composition confound) shows PEER + TEXT at 0.753 and c_score_align at 0.741, a difference of +0.011 [−0.012, 0.038], not significant. On the long pool the stratified gap is +0.016 [−0.010, 0.041]. On R_A the gap is −0.036 [−0.113, 0.030], favouring c_score_align. The TEXT layers add no significant discriminative power over c_score_align alone once the stratum-composition confound is removed.

Mechanism predictions

Four pre-registered mechanism predictions were tested on E:

P1 (PEER catches COVERAGE errors that TEXT misses, and vice versa). At matched false-alarm rate 0.10 with fractional tie-breaking: PEER has higher recall than TEXT on COVERAGE errors (+0.093 [0.046, 0.147]), and TEXT has higher recall on PEER_ENDORSED errors (+0.103 [0.051, 0.171]). The crossover exists, but the fused PEER + TEXT recall is not higher than c_score_align

alone on any group except COVERAGE. Verdict: INCONCLUSIVE.

P2 (PEER and TEXT error rankings are uncorrelated). Spearman correlation between PEER and TEXT among errors is 0.294 [0.224, 0.365]. This is moderate, not near zero. The gain from fusion over the best single signal (PEER) is only +0.042 AUROC, and the share from PEER-endorsed errors is −0.200, meaning TEXT hurts on items PEER already handles. Verdict: REFUTED.

P3 (PEER + TEXT advantage grows with sentence complexity). The logistic interaction term $z(\text{score}) \cdot z(\text{words})$ is −0.023 [−0.228, 0.185] for PEER + TEXT, consistent with no interaction. The diff-in-Δ (long minus CTRL) of PEER + TEXT minus the local judge is +0.146 [0.020, 0.269], but this reflects the judge degrading on long sentences, not the fusion improving. Verdict: INCONCLUSIVE.

P4 (PEER + TEXT is more invariant to meaning-preserving rewrites than judges). On the screen pools, the fused score's RENAME false-alarm rate is 0.385 and the local judge's is 0.683. The fused score is better than judges but far worse than FOL-Triage's 0.013. The `c_score_align` component drives most of the fused FA under RENAME (0.859). Verdict: REFUTED.

Rewrite false alarms (screen, CORRECT bases)

Rewrite	n	Fused	<code>c_score_align</code>	NF g	L2-bow	L3	Local judge
RENAME	104	0.385	0.859	0.074	0.216	0.168	0.683
REORDER	27	0.222	0.178	0.037	0.044	0.444	0.000
DEMORGAN	26	0.154	0.185	0.038	0.046	0.231	0.231
CONTRAPOSITIVE	28	0.107	0.063	0.237	0.075	0.188	0.214

The NF g_score has the lowest RENAME false-alarm rate (0.074) of any consensus-based signal, confirming the value of name-free matching. However, its CONTRAPOSITIVE FA (0.237) is higher than `c_score_align`'s (0.063).

Error typing

On R_A errors with 1-2 typed repair operators (n = 453), the top-code accuracy of the medoid-ops typed repair is 0.371 (codable-only 0.561, n = 262). Chance level is 0.382. The local judge's type-classification accuracy is 0.022 (10/453). Error typing from automated metrics remains unreliable.

System-level correlation

Kendall τ -b of each metric vs the R_AB system error rate across 13 system × variant rows: PEER + TEXT 0.718, TEXT alone 0.769, PEER alone 0.692, local judge 0.667. All metrics track system quality at the aggregate level.

Contamination on E

The local Qwen3-8B judge shows a disguise effect in the unexpected direction: disguised AUROC (0.712) exceeds original AUROC (0.648) by +0.064 [0.032, 0.098]. This is likely because nonce disguise forces the model to attend to structural cues rather than relying on surface familiarity. The DiD test (comparing the disguise effect on track H vs track L) yields +0.045 [−0.073, 0.161], with the CI including zero. No contamination is detected. The PEER signal has no contamination channel, because it uses z3 equivalence (no text or names from the benchmark enter the scoring).

Coverage and cost

Of 8,507 rows, 7,421 (87.2%) are parseable and scored; 1,086 (12.8%) are unparseable (scored as maximum error probability). 38 rows have no peer translations available. The L3 questionnaire completed on all parseable rows. Mean CPU time per row: PEER pairs 0.026 s, NF pairs 0.208 s, TEXT 0.220 s. Mean cost per row for PEER + TEXT (L3 share): \$0.000018. The local judge costs \$0.000040 per row.

2.3 Evaluation 1: Re-evaluation under alternative label regimes

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The iteration-1 experiments used solver labels (R_SOLVER): a formula is CORRECT if z3 proves it equivalent to the audited gold, ERROR otherwise. This is conservative: formulas that are semantically faithful but use different vocabulary or granularity are counted as errors. The parallel run adjudicated a panel of labels (R_ADJ) on the same items, where three LLM judges vote on faithfulness given only the sentence and candidate. This evaluation re-scores all metrics under 13 label regimes on a common item set (588 items shared across experiments A, C, D, and E).

Rederivation audit

All 57 reviewer-audited quantities from the iteration-1 report reproduce exactly under independent recomputation (57/57 MATCH, tolerance ≤ 0.01 for AUROCs, ≤ 0.005 for bootstrap replicas). The reviewer's own RNG seed replicas match to ≤ 0.0005 .

Regime AUROC matrix (common set, selected metrics)

Metric	R_SOLVER	R_ADJ_AB	R_ADJ_A	R_ADJ_ALL
PT (PEER + TEXT)	0.871	0.877	0.932	0.906
fused_H (FOL-Triage)	0.770	0.868	0.958	0.866
c_score	0.849	0.779	0.786	0.831
L2-bow	0.732	0.815	0.917	0.810
L3	0.761	0.718	0.813	0.751
judge_cheap_disg	0.782	0.772	0.751	0.816
judge_cheap_orig	0.754	0.797	0.858	0.785
sc5_cheap	0.713	0.703	0.676	0.737
rt_nli_min	0.709	0.749	0.731	0.760

Under R_ADJ_AB, PEER + TEXT (0.877) leads, followed by FOL-Triage fused (0.868) and L2-bow (0.815). The c_score drops from 0.849 (solver) to 0.779 (R_ADJ_AB), because the adjudicated labels reclassify many vocabulary-mismatch items as CORRECT, removing the easy wins for the consensus signal. Under R_ADJ_A (tier-A only, 75 common items), FOL-Triage reaches 0.958 and PEER + TEXT 0.932, both well above the judges.

Label regime shifts

On the 161 items shared between R_SOLVER and R_ADJ_AB (36 label changes), the label-only shift is:

Metric	Shift	95% CI	Direction
fused_H	+0.080	[0.003, 0.171]	UP
L2-bow	+0.102	[0.018, 0.192]	UP

known by construction. All mutants are z3-verified: the DOWN mutant is z3-non-equivalent to the base and the matched CONTROL is z3-equivalent.

Statistic	Value
Base formulas	300
Total mutants (DOWN + UP)	4,234
Matched controls	868
Total suite rows	5,102
Complete DOWN/UP pairs	1,354
z3 verification failures	0
Operator types	12 (NEG, REV, QUANT, RESTR, CONN, MOVE, DROP, ADD, SWAP, BIND, SCOPE, UNGLUE)
Cost	\$0.41

Each perturbation row carries a known label (ERROR for DOWN mutants, CORRECT for controls) and a typed operator, enabling per-type sensitivity analysis without relying on automated error classification. The suite is fully deterministic and re-runnable.

2.5 Iteration 2 Summary

What iteration 2 established:

1. PEER + TEXT achieves AUROC 0.790 [0.75, 0.82] on the held-out dataset E (R_AB pooled, n = 2686), beating the local judge by +0.078 [0.043, 0.112] (DeLong p < 0.0001). On long sentences (L25 + L20 + EXC, n = 2300) the gap widens to +0.115 [0.075, 0.157]. The flash-lite judge bar is UNTESTABLE due to API key exhaustion (only 20 rows scored).
1. Fusion does not significantly beat c_score_align alone. The stratified AUROC difference (PEER + TEXT minus c_score_align) is +0.011 [-0.012, 0.038] on R_AB pooled. The TEXT layers add coverage on COVERAGE-type errors (+0.093 recall at matched FA) but hurt on PEER-endorsed errors (-0.200 share). The mechanism prediction P2 (uncorrelated error rankings) is REFUTED: Spearman(PEER, TEXT) = 0.294.
1. The name-free consensus variant (NF-anchored) failed the pre-registered screen gate due to a threshold-tie artefact. Post-hoc fractional scoring shows it passes both gates (RENAME FA 0.074, ROLE_PERMUTE recall 0.771) and achieves AUROC 0.759 on E, 0.031 below PEER + TEXT. The NF approach has the lowest RENAME false-alarm rate (0.074) of any consensus signal, confirming the value of vocabulary-independent matching, but was not used in the primary analysis.
1. Under adjudicated labels (R_ADJ_AB), the metric ranking changes. FOL-Triage fused rises from 0.770 (solver) to 0.868 (adjudicated). The c_score drops from 0.849 to 0.779. This is because adjudicated labels reclassify vocabulary-mismatch items as CORRECT, removing the consensus signal's easy wins. PEER + TEXT (0.877) still leads under adjudicated labels.
1. The frontier judge advantage is large under adjudicated labels (+0.166 AUROC over the cheap judge under R_ADJ_AB, p < 0.001), reversing the non-significant iteration-1 finding

under solver labels. Solver labels masked the frontier judge's advantage.

1. All 57 reviewer-audited numbers from iteration 1 reproduce exactly (57/57 MATCH).

1. The PERTURB suite (4,234 typed mutants + 868 controls, 12 operator types, all z3-verified) is complete and ready for per-type sensitivity analysis. R_COMP candidate generation is blocked on the OpenRouter key.

What remains:

(a) Run the flash-lite judge on E when the OpenRouter key resets, to test the pre-registered bar. (b) Complete R_COMP candidate generation and labelling. (c) Score all metrics on the PERTURB suite for per-type sensitivity under known labels. (d) Re-fit the PEER + TEXT fusion under R_ADJ labels (the current fusion was fitted under solver labels, which penalise it). (e) Test whether a name-free fusion (NF-anchored + TEXT) closes the 0.031 gap with the aligner-based fusion when the aligner confound is removed.

Related Work

NL-FOL datasets and their quality. FOLIO [1] provides 1,435 FOL-annotated NLI problems. MALLS [2] uses GPT-4 to generate and verify NL-FOL pairs at scale. Brunello et al. [4] found that approximately 42% of entries in both datasets contain incorrect FOL formalisations, with additional ambiguity rates of 17.8% (FOLIO) and 51% (MALLS). Their LLM-assisted re-labelling framework reduces human effort by $5\text{--}15\times$ compared to exhaustive review. Logic-LM [10] prompts LLMs for FOL and calls a prover, releasing outputs that we use as our evaluation screen.

FOL closeness metrics. Thatikonda et al. [5] evaluate n-gram, graph, embedding, and LLM-based metrics on controlled FOL perturbations, finding that no single metric is sensitive to all perturbation types. Smatch + + is the most sensitive to quantifier changes, while BL-score responds most to negation. Their work measures sensitivity to synthetic perturbations; ours measures faithfulness prediction on real LLM outputs with noisy labels.

NL-FOL error taxonomies. Thatikonda et al. [6] propose a verification pipeline with a 28-category error taxonomy. Brunello et al. [4] classify errors into semantic and structural categories. Our typed repair operators (NEG, REV, QUANT, RESTR, CONN, MOVE, DROP, ADD, SWAP, BIND, SCOPE, UNGLUE) follow the same tradition but are derived from z3 equivalence-class differences rather than manual annotation.

Factuality metrics. FRANK [11] defines a typology of factuality errors in abstractive summarisation and evaluates correlation of automatic metrics with human judgments. QuestEval [13] uses question generation and answering to assess factual consistency. QAGS [12] similarly generates questions from a summary and checks whether the source answers them the same way. Our L3 questionnaire layer is closest to this family: it generates structured questions from the text and checks the formula's answers via z3, rather than using another LLM.

Round-trip and self-consistency. Amrollahi et al. [7] propose verbalising a formal statement back to natural language and checking consistency, reporting high accuracy on mathematical autoformalization. Li et al. [8] use symbolic equivalence and semantic consistency for Lean/Isabelle autoformalization. Our round-trip baseline follows the same pattern but achieves only AUROC

0.710 on NL-FOL, below the LLM judge.

Monotonicity and polarity. SyGNS [14] tests monotonicity reasoning in NLI models. Udep2Mono [15] computes monotonicity profiles from Universal Dependencies. Our L2 role-aware layer attempted to use monotonicity-aware comparisons but achieved AUROC 0.557, a negative result.

Counter-interpretation methods. CLOVER [9] generates counter-interpretations to test logical validity via an LLM oracle. Our consensus approach is related in spirit: it uses multiple independent translations as implicit counter-proposals, but relies on z3 equivalence rather than LLM-generated counterexamples.

Cross-check consistency. SAC3 [17] detects hallucinations in black-box LLMs by sampling diverse perturbations of a query and checking consistency across answers. Our cross-system consensus metric uses a similar principle: disagreement among independent translations signals error. It operates on FOL equivalence rather than textual similarity, and uses multiple model families rather than perturbations of a single model. The Δ AUROC of cross-family consensus over single-model self-consistency is +0.140 in our setting, consistent with SAC3's finding that semantic-aware cross-checking outperforms naive self-consistency.

Automated alignment for autoformalization. FormalAlign [18] trains a model on both autoformalization and alignment scoring via a dual contrastive-generative loss, targeting Lean/Isabelle autoformalization. It is the closest published gold-free alignment scorer for formal languages. Our approach differs in that we use no learned alignment model: the consensus signal relies on z3 equivalence among independent translations, and the text-based signal uses deterministic checks and a prompted questionnaire. FormalAlign targets mathematical theorem proving, where the target language has a type checker; NL-FOL has no such verifier, making gold-free evaluation harder.

Vacuity detection. Kupferman and Vardi [16] formalise vacuity in temporal logic model checking. Our L1 lint layer includes a vacuity check (detecting formulas where a subformula can be replaced by its negation without changing the truth value), adapted from their framework.

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